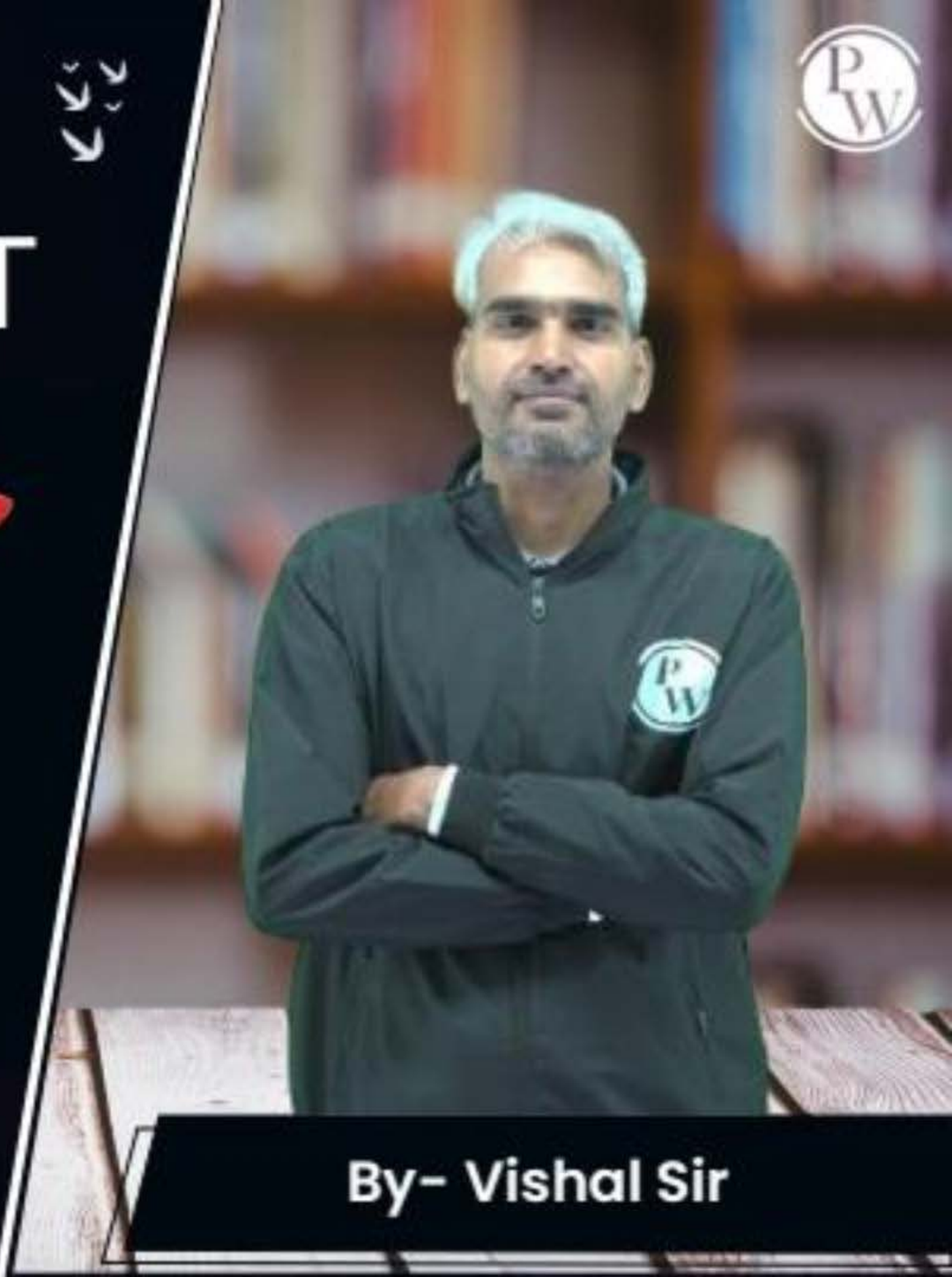
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 17



By- Vishal Sir

Recap of Previous Lecture



Topic Surjective (onto) function

Topic Number of onto functions

Topic Bijective function

Slide

Topics to be Covered



- Topic: Inverse of a function
- Topic: Identity function
- Topic: Constant function
- Topic: Identical function
- Topic: Function composition

Slide



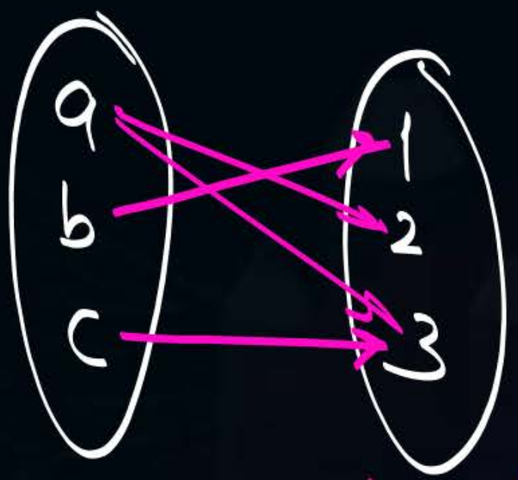
Topic : Function

A relation from set A to set B is called a function from set A to set B if and only if

Every element of set A relates with exactly one element of set B

{ Function from set A to set B is also a relation from set A to set B }

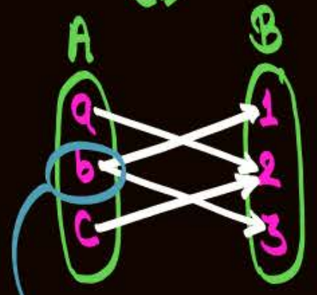
{ Every function is a relation, but every relation need not be a function }



It is a valid relation from A to B but is not a function from A to B

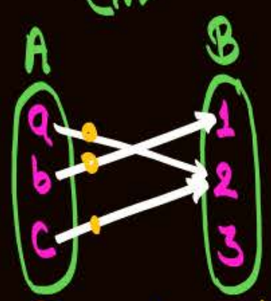
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(i)



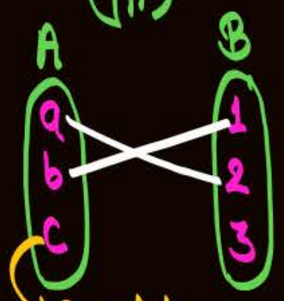
element 'b' of set A relates with two elements of set B
∴ Not a function from A to B

(ii)



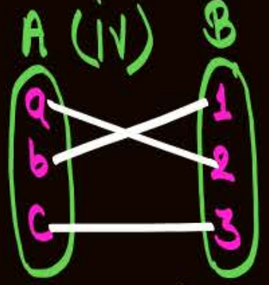
Every element of set A relates with exactly ONE element of set B
∴ It is a function from A to B

(iii)



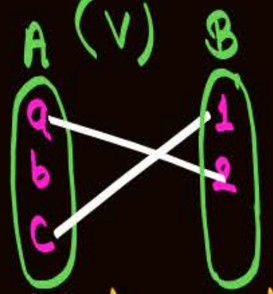
'c' relates with '0' elements of set B.
∴ Not a function from A to B

(iv)



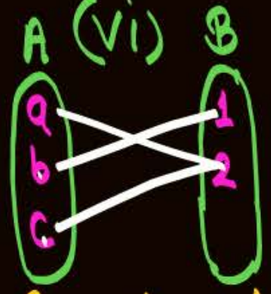
Every element of set A relates with exactly ONE element of set B
∴ It is a function from A to B

(v)



'b' does not relate with any element of set B
∴ Not a function from A to B

(vi)



Every element of set A relates with exactly ONE element of set B
∴ It is a function from A to B

(i) Neither a function from A to B nor a function from B to A

(ii) function from A to B but not a function from B to A

(iii) Neither a function from A to B nor a function from B to A

(iv) Function from A to B as well as function from B to A.

(v) Not a function from A to B, but a function from B to A

(vi) function from A to B, but not a function from B to A



Topic : Function

- A function 'f' from set A to B is denoted by function $f: A \longrightarrow B$

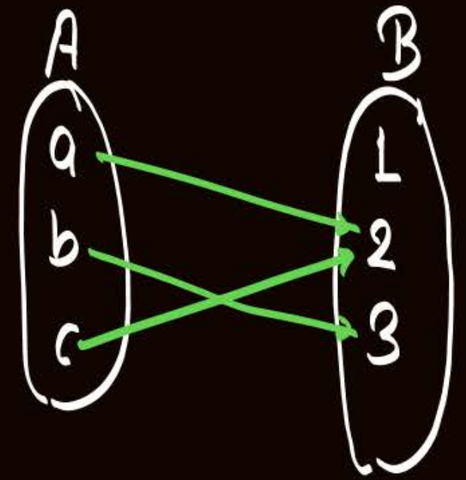
- Let $f: A \to B$ is a function, then

  Set A is called domain of the function

  & Set B is called Co-domain of the function

Slide

eg.



function $f: A \to B$

Domain = $\{a, b, c\}$

Co-domain = $\{1, 2, 3\}$

In a function
- it is not necessary for every element of the co-domain to be mapped by at least one element of the domain

eg. '1' is not mapped by any element of domain



Topic : Range of a function

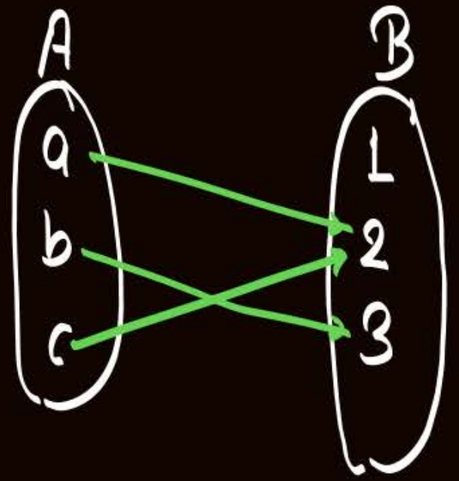
It is a set of all the elements of the co-domain that are mapped by at least one element of domain

In general

Range of function $\subseteq$ Co-domain of function

Slide

eg.



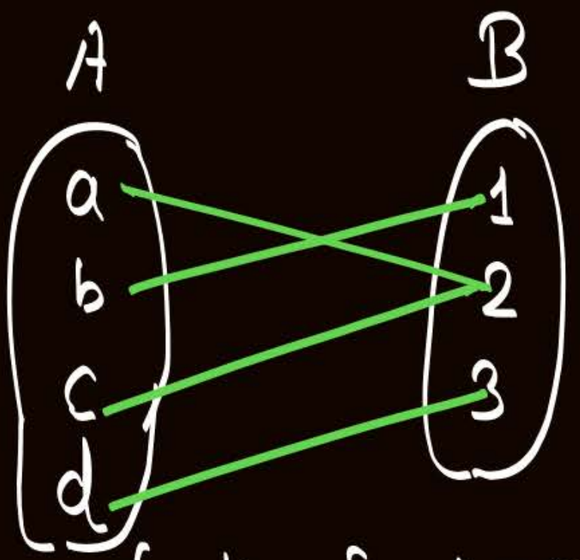
function $f_1: A \to B$

Domain = {a, b, c}

Co-domain = {1, 2, 3}

Range = {2, 3}

Range $\subset$ Co-domain



function $f_2: A \to B$

Domain = {a, b, c, d}

Co-domain = {1, 2, 3}

Range = {1, 2, 3}

Range = Co-domain

Note: ① Function must be defined for every element of the domain.

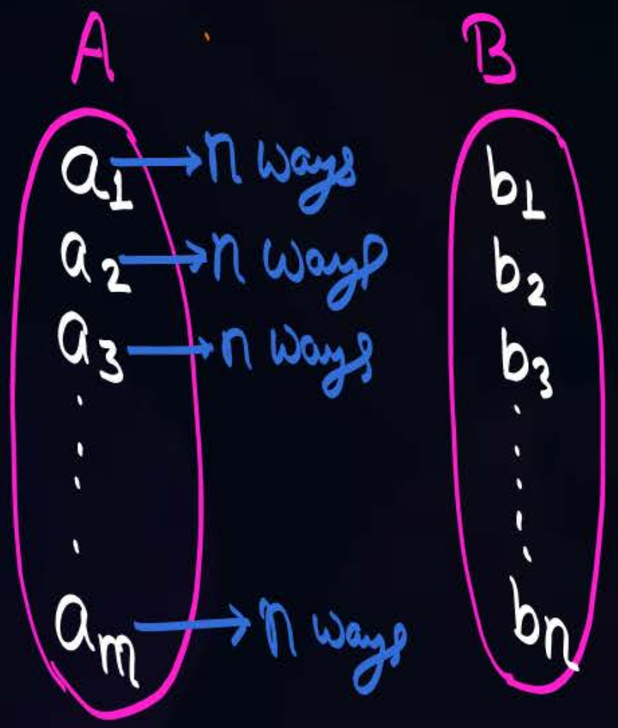
② The result of the function on the input from its domain can not acquire a value which is not present in Co-domain.



Topic : Total number of functions

let $|A| = m$ & $|B| = n$, then

total Number of functions possible from set A to set B = n^m = $(\text{Size of Co-domain})^{\text{Size of domain}}$



$$= \underbrace{n * n * n * \ldots * n}_{m \text{ times}}$$

$$= n^m$$

Slide

Note:- A function from set A to set A itself is called a function on set A.

Note:- if $|A| = n$, then

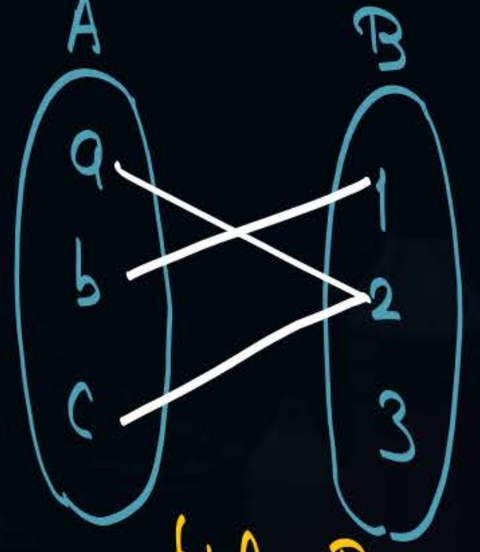
Total number of functions possible on set A : n^n

Topic : Injective (one-one) function

A function $f: A \to B$ is called an injective (one-one) function only if distinct elements of domain have distinct images in co-domain

i.e., if $a \neq b$, then $f(a) \neq f(b)$

i.e., a & b are distinct → then → images of 'a' & 'b' must be distinct



$f: A \to B$

f(a) is called image of element 'a' w.r.t function 'f'

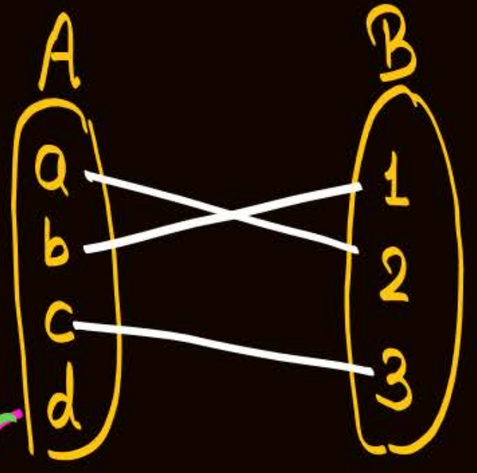
{'a' is called Pre-image of f(a)}

it is a function but not a one-one function

as $a \neq c$ but $f(a) = f(c) = 2$

Two pre-images of element '2'

Slide



if there is no image of 'd' then it will not be a function

if there is an image of 'd' then it will be same as one of a, b or c

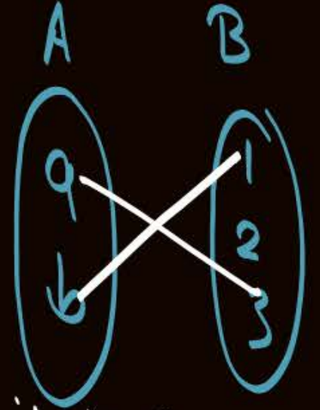
∴ it will not be one-one function

If $|A| > |B|$, then

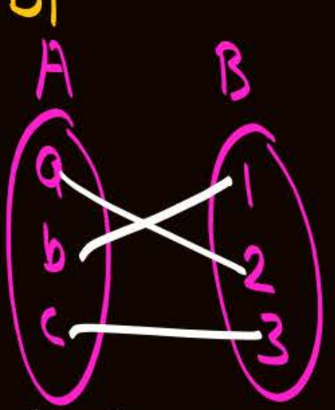
One-one function is not possible from set A to set B

A one-one function from set A to set B is possible only if,

$$|A| \leq |B|$$



it is one-one function



it is one-one function

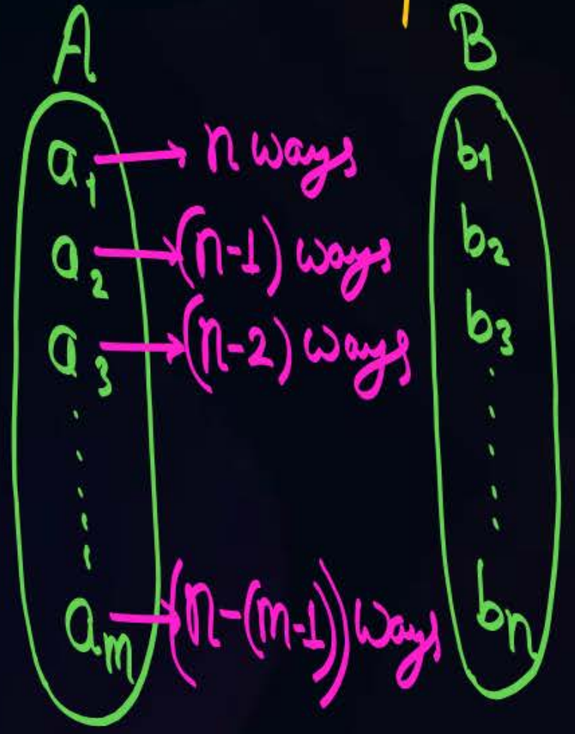
Note: In a one-one function, every element of Co-domain will have at most one pre-image

Topic : Number of one-one function

Let $|A| = m$ & $|B| = n$ such that $m \le n$,

then,

total number of one-one functions possible from A to B $= {}^{n}P_{m} = \frac{n!}{(n-m)!}$



$= n*(n-1)*(n-2)*\cdots\cdots(n-m+1) * \frac{(n-m)*(n-m-1)*\cdots 3\times 2\times 1}{(n-m)*(n-m-1)*\cdots 3\times 2\times 1}$

$= \frac{n!}{(n-m)!} = {}^{n}P_{m}$

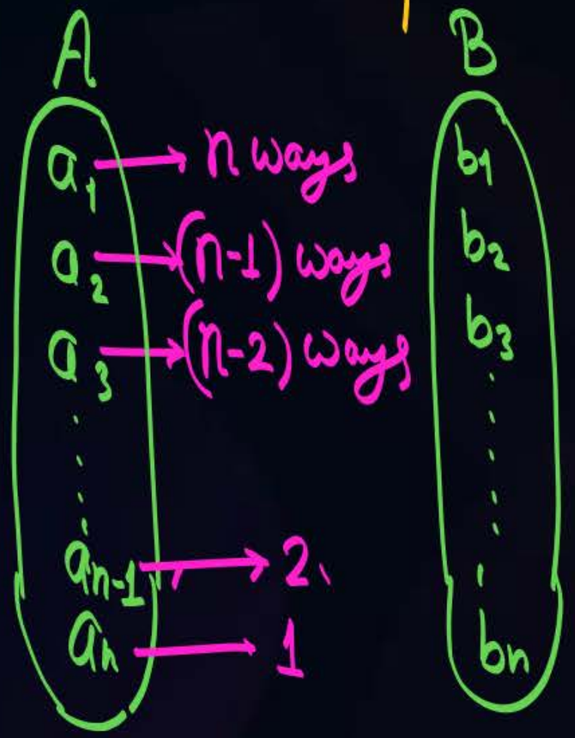
Slide

Topic : Number of one-one function

let $|A| = |B| = n$,

then,

total number of one-one functions possible from A to B = $n!$ $\Rightarrow$ $\boxed{{}^{n}P_{n} = \frac{n!}{(n-n)!} = n!}$



$= n*(n-1)*(n-2)*\cdots*3*2*1$

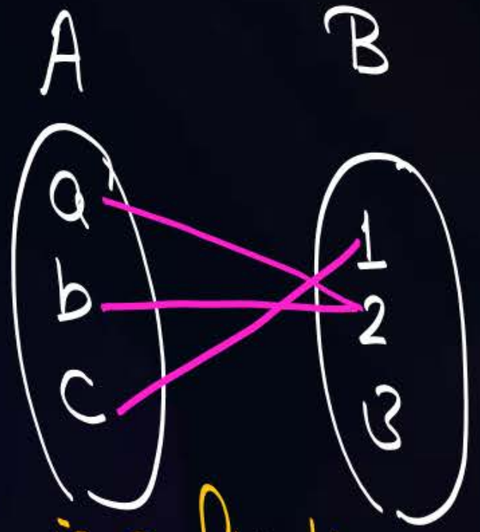
$= n!$

Slide



Topic : Surjective (onto) function

A function $f: A \to B$ is called surjective (onto function) if and only if every element of Co-domain is mapped by at least one element of domain

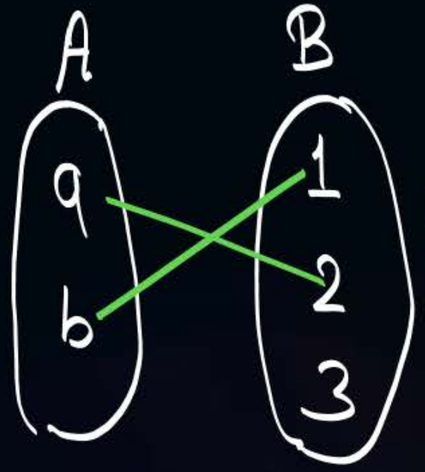


element '3' is not mapped by any element of domain

it is a function, but not an 'onto' function

Slide

Topic : Surjective (onto) function



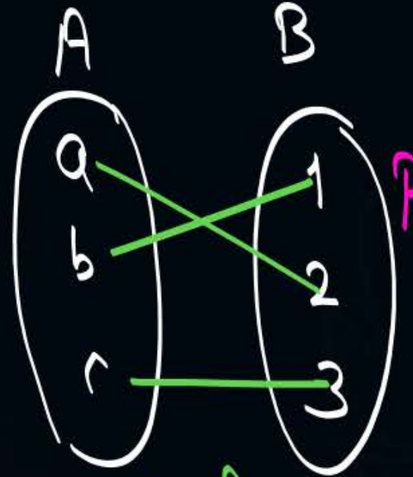
Range ⊂ Codomain

$f: A \to B$

it is a function, but it can not be onto, if we try to map '3' with one of the elements of set A, then it will not remain a function

An onto function is possible from set A to set B, only if $|A| \geq |B|$

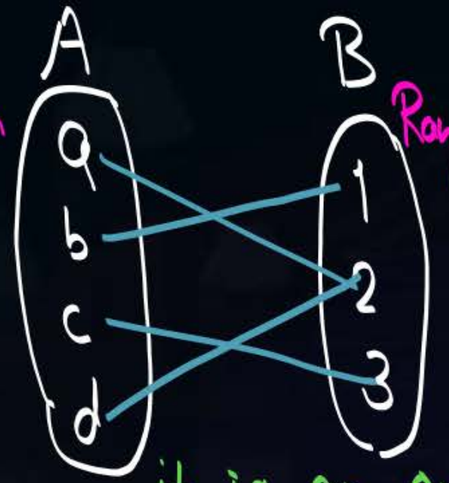
$|A| = |B|$



Range = Co-domain

onto function
{Also one-one function}

$|A| > |B|$



Range = Co-domain

it is an onto-function (but not one-one function)

Slide

① In an onto function.

Range of function = Co-domain of function

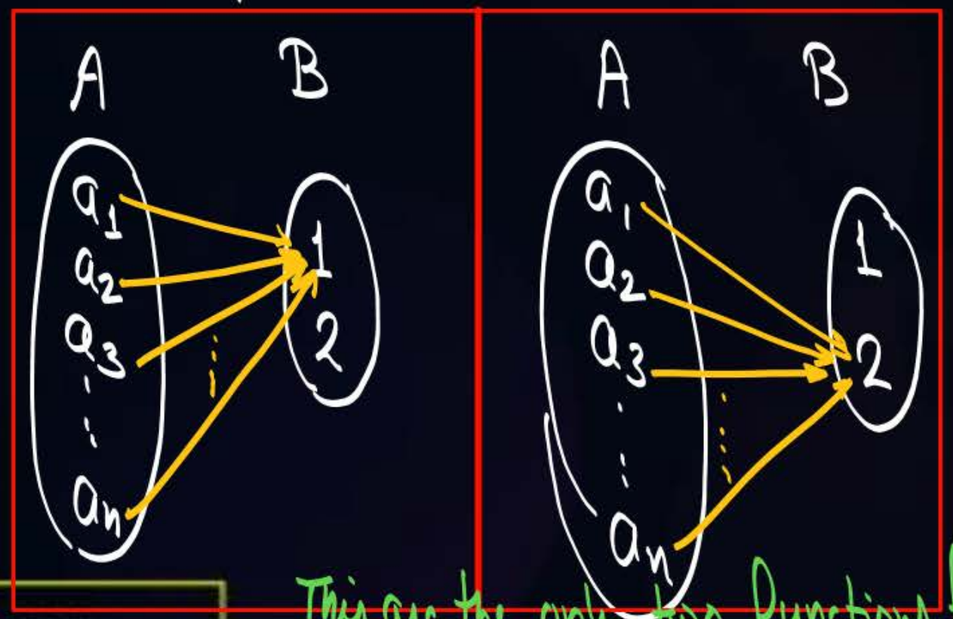
② In an onto function every element of Co-domain will have at least one Pre-image

Topic : Number of onto functions

Case ① let $|A| = n$ & $|B| = 2$, $(n \geq 2)$

Number of onto-functions Possible from A to B = (Total no. of functions from A to B) − (No. of functions which are not onto)

$= 2^n - 2$



This are the only two functions from Set A to Set B which are not "onto"

Slide



Topic : Number of onto functions

Case ① let $|A| = n$ & $|B| = 2$, $(n \geq 2)$

Number of onto-functions Possible from A to B $= 2^n - 2$

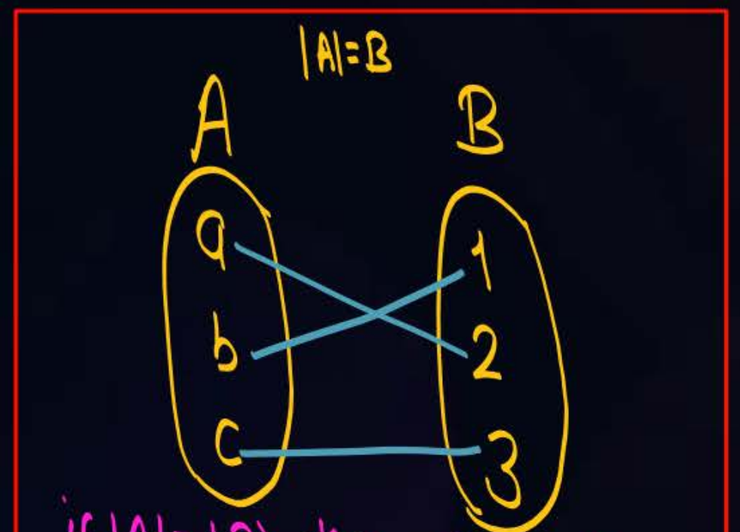
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Topic : Number of onto functions

Case ②

If $|A| = |B| = n$.

Number of onto functions Possible from A to B = Number of one-one functions from A to B $= {}^{n}P_{n} = n!$



if $|A| = |B|$, then every one-one function from A to B will also be "onto".
& Every onto function from A to B will also be one-one

Slide



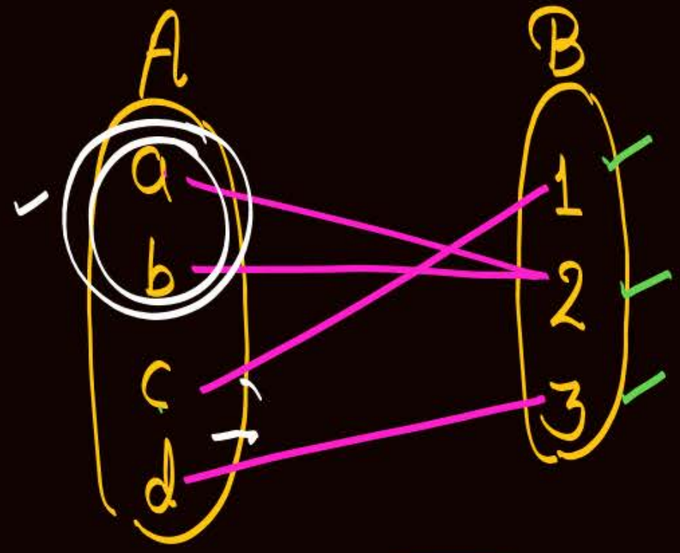
Topic : Number of onto functions

Q: Let $|A| = 4$ & $|B| = 3$, (n) ($n-1$)

then how many Onto functions are possible from set A to set B $= {}^4C_2 * 3!$

$= 6 * 6 = 36$

Slide



In any onto function from A to B
there will be exactly two elements in
set A which will map to the same element of set B.

Every other element of set A will
map with distinct elements of set B

∴ No. of onto functions from A to B $= {}^4C_2 * 3 * 2!$

$= {}^4C_2 * 3!$



Topic : Number of onto functions

Case ③ $|A| = n$ & $|B| = (n-1)$

i.e. $\boxed{|A| = |B| + 1}$

Number of onto functions Possible from set A to set B $= {}^{n}C_{2} * (n-1)!$

Slide

Topic : Number of onto functions

Case ④ let $|A| = m$ & $|B| = n$ s.t. $m \geq n$

No. of onto functions Possible from set A to set B = Total no. of functions from set A to set B − (No. of functions which are not onto)

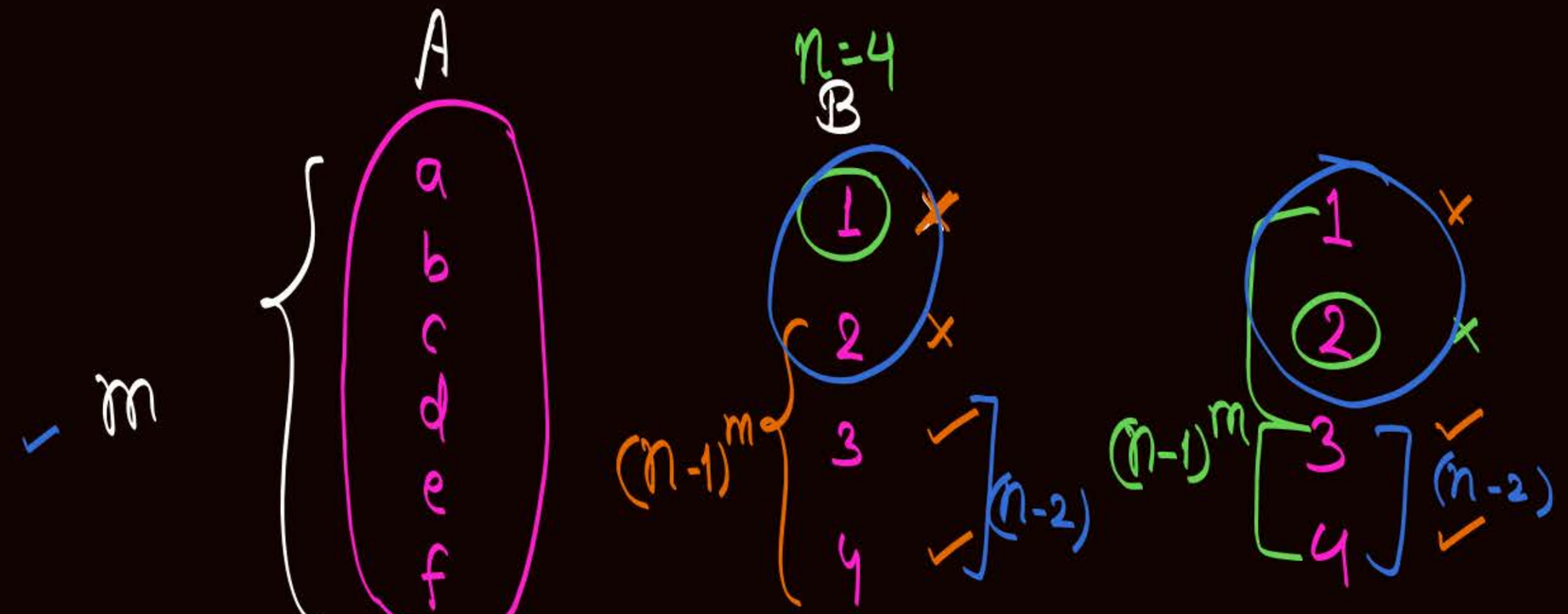
(No. of functions in which one elemental co-domain is not mapped by any element of domain + No. of functions in which two elements are not mapped + ... + No. of functions in which (n−1) elements are not mapped)

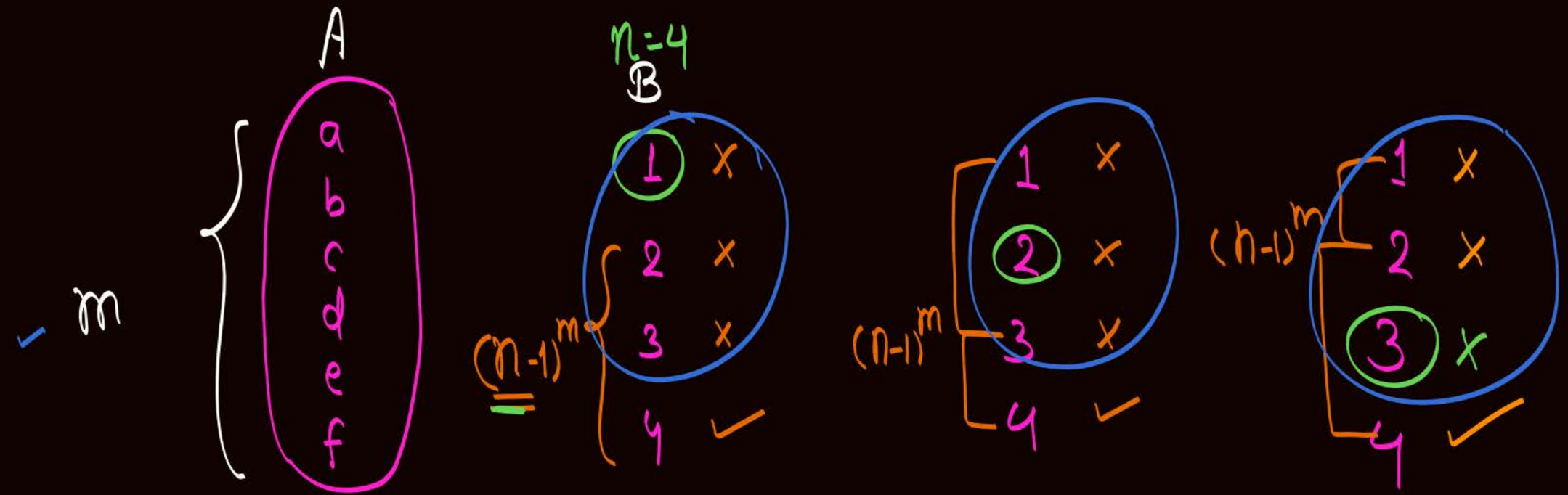
$$= n^m - {}^nC_1 * (n-1)^m + {}^nC_2 * (n-2)^m - {}^nC_3 * (n-3)^m + \cdots + (-1)^{n-1} {}^nC_{n-1} (n-(n-1))^m$$

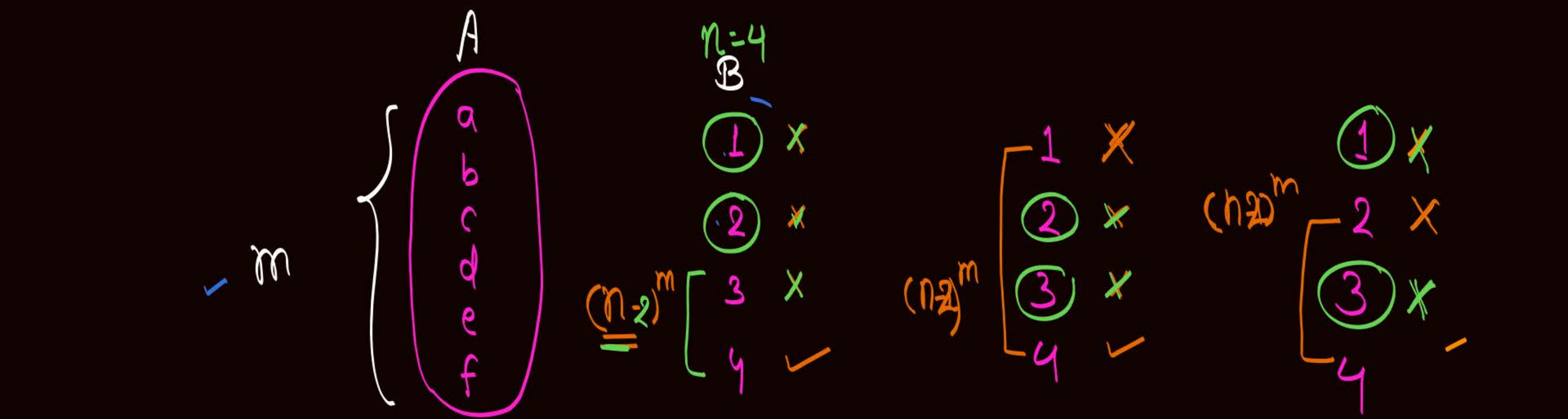
$$= {}^nC_0 \cdot (n-0)^m - {}^nC_1 * (n-1)^m + {}^nC_2 (n-2)^m - {}^nC_3 \cdot (n-3)^m + \cdots + (-1)^{n-1} \cdot {}^nC_{n-1} (n-(n-1))^m$$

$$= \sum_{i=0}^{n-1} (-1)^i \cdot {}^nC_i * (n-i)^m$$

Slide


A
n=4
B
m
a
b
c
d
e
f
1
2
3
4
(n-1)^m
(n-2)
(n-1)^m
(n-2)


A
m
a
b
c
d
e
f
n=4
B
1
x
2
x
3
x
4
$(n-1)^m$
$(n-1)^m$
$(n-1)^m$


A
a
b
c
d
e
f
m
n=4
B
1
2
3
4
x
(n-2)^m
(n-2)^m
(n-2)^m

Case		No. of onto functions
①	$\lvert A\rvert = n$ & $\lvert B\rvert = 2$	$2^n - 2$
②	$\lvert A\rvert = \lvert B\rvert = n$	$n!$
③	$\lvert A\rvert = n$ & $\lvert B\rvert = n-1$	${}^{n}C_2 \times (n-1)!$
④	$\lvert A\rvert = m$ & $\lvert B\rvert = n$, $m \geq n$	$\sum_{i=0}^{n-1} (-1)^i \cdot {}^{n}C_i \cdot (n-i)^m$



Topic : Bijective Function

- A function $f: A \to B$ is called a bijective function if and only if, (1) $f: A \to B$ is one-one (Injective) & (2) $f: A \to B$ is onto (Surjective)

- A function is bijective iff it is one-one as well as onto (i.e. $|A| \leq |B|$) (i.e., $|A| \geq |B|$)

- A bijective function from set A to set B is possible only if $|A| = |B|$

- If $|A| = |B| = n$, then Number of bijective functions possible from A to B = $n!$

Slide

Note: If there exist a bijective function from set A to set B (or from set B to set A), then set A and set B are said to have one-one correspondance

$\therefore$ One-one Correspondance $\Longrightarrow$ One-one as well as onto

- if One-one Correspondance exists between set A & set B, then $|A| = |B|$



Topic : Inverse of a Function

- Inverse of a function 'f' can be defined if and only if function 'f' is a bijective function.

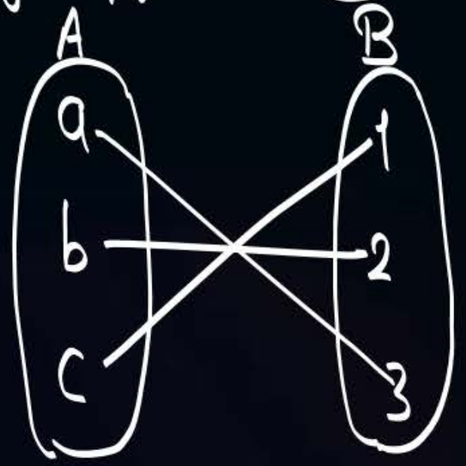
- If $f: A \to B$ is a bijective function, then inverse relation of function 'f' will also be a function denoted by f^{-1} and f^{-1} is called inverse function of function f.

- If 'f' is a bijective function from A to B, then f^{-1} will be a function from set B to set A.

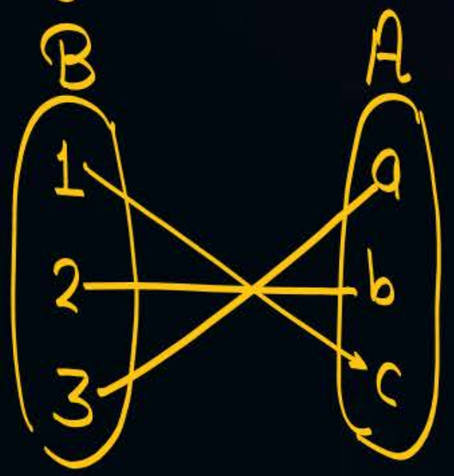
Slide

Topic : Inverse of a Function

eg. $f : A \longrightarrow B$



$f^{-1} : B \to A$



$f(a) = 3$
$f(b) = 2$
$f(c) = 1$

$\Longrightarrow$

$f^{-1}(3) = a$
$f^{-1}(2) = b$
$f^{-1}(1) = c$

$\Rightarrow$ i.e. if $f(x) = y$ then, $f^{-1}(y) = x$

Slide

eg: Consider the following bijective function

$f(x) = 2x + 3$

Obtain the inverse function w.r.t. function $f(x)$

i.e., $f^{-1}(x) = ?$

Soln:-

Let $y = f(x)$

$\therefore f^{-1}(y) = x$

$f^{-1}(y)$ must be a function of y $\therefore$ replace 'x' in terms of y.

Let $y = f(x)$

i.e. $y = 2x + 3$

$\therefore x = \left(\frac{y-3}{2}\right)$

$\Longrightarrow \therefore f^{-1}(y) = x$

$f^{-1}(y) = \frac{y-3}{2}$

$\Rightarrow$ y is just a variable $\therefore \therefore f^{-1}(x) = \left(\frac{x-3}{2}\right)$

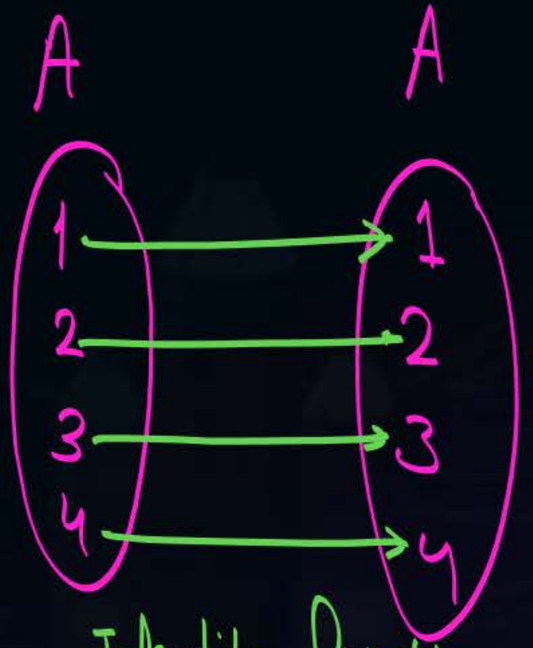


Topic : Identity Function

A function 'f' from set A to set A {i.e., $f: A \to A$} is called an identity function if and only if

$$f(x) = x, \quad \forall x \in A$$

- Identity function on set A is denoted by I_A
- Every identity function is a bijective function.
- Inverse of identity function is identity function itself



Identity function on set A

Slide

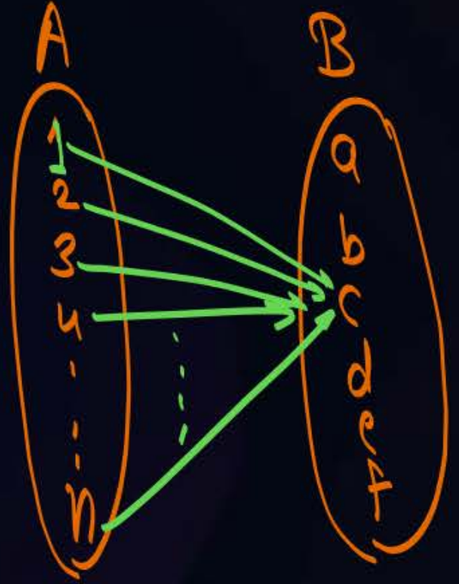


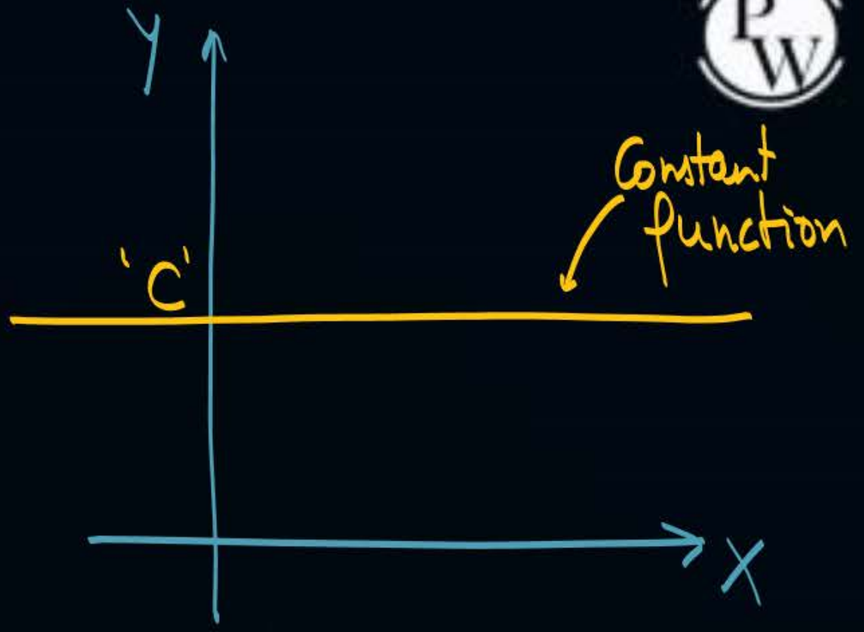
Topic : Constant Function

A function $f: A \to B$ is called a constant function if,

$$f(x) = c, \quad \forall x \in A$$

When $c \in B$





Slide



Topic : Identical function

Equal Function

Two functions f & g are said to be identical if and only if

(i) domain of f = domain of g

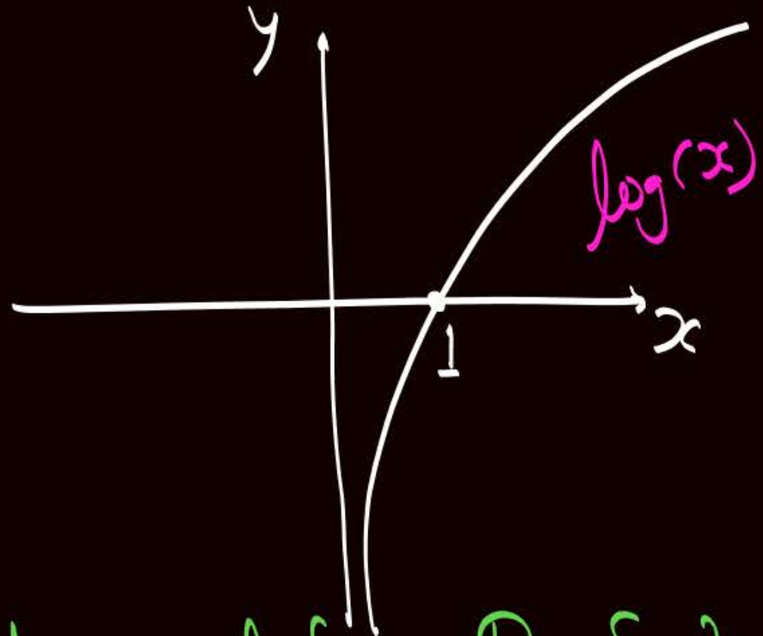
& (ii) Range of f = Range of function g

& (iii) $f(x) = g(x) \quad \forall x \in$ domain

Slide

Q: Let $f(x) = \log(x^2)$ & $g(x) = 2\log x$

Check whether $f(x)$ & $g(x)$ are identical functions or not?



$f(x)$ & $g(x)$ are not identical

domain of $f(x) = R - \{0\}$ $\neq$ domain of $g(x) = R^+$ (Set of all +ve real numbers)

Range of $f(x) = R$ $=$ Range of $g(x) = R$



Topic : Function comopostition

Let f & g are two functions, then Composite function of function f and g is denoted by "fog"

In general $fog \neq gof$

{i.e. function composition is not commutative}

- $(fog)(x) = f(g(x))$
- $(gof)(x) = g(f(x))$

Slide

$$\underset{f_1}{((f \circ g)} \underset{f_2}{\circ h})(x) \qquad \underset{f_1}{(f \circ} \underset{f_2}{(g \circ h)})(x)$$

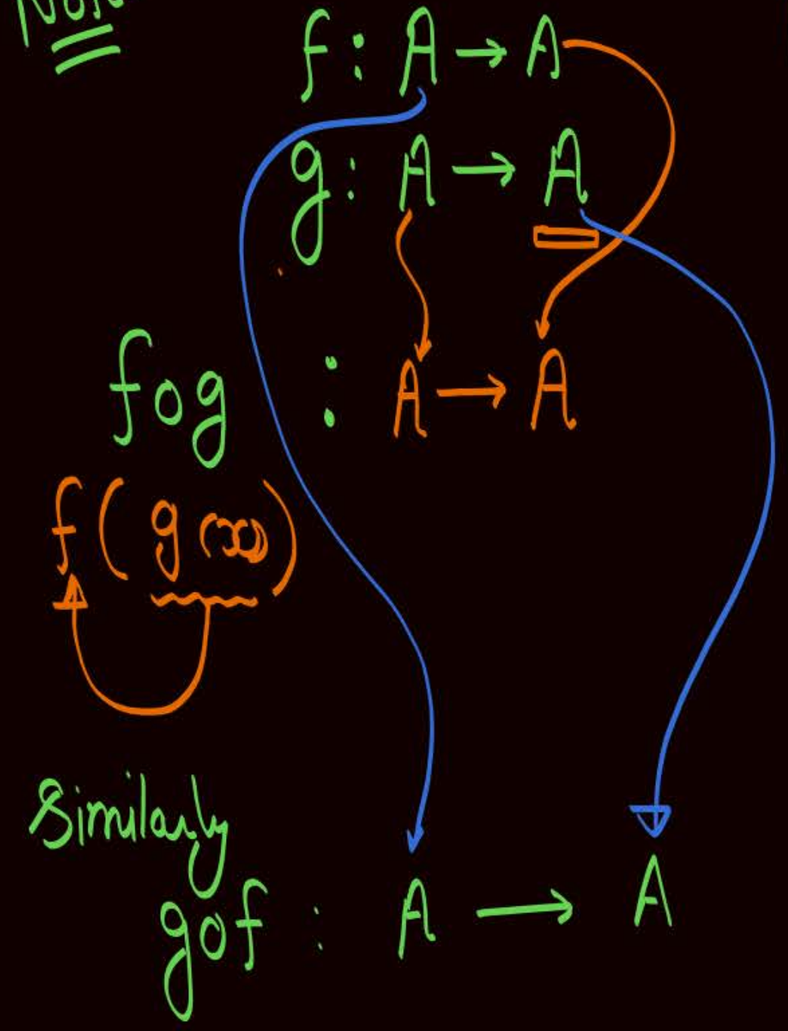
$$(f \circ g)\,(h(x)) \qquad f\,((g \circ h)(x))$$

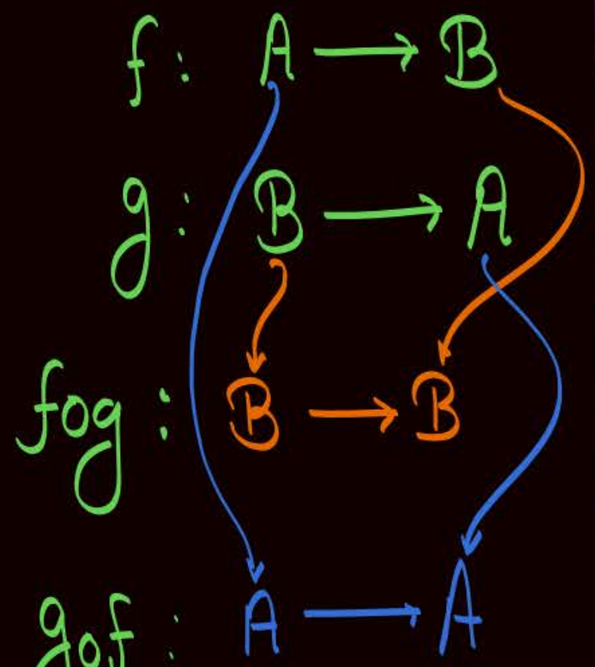
$$f\,(g(h(x))) \quad = \quad f\,(g(h(x)))$$

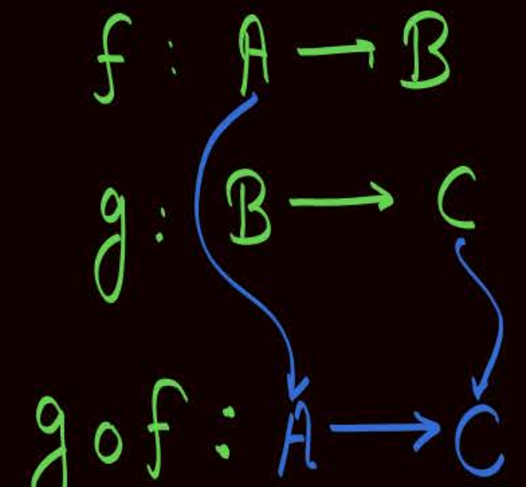
$\therefore$ function composition is associative

$$\boxed{f_1 \circ (f_2 \circ f_3) = (f_1 \circ f_2) \circ f_3}$$

Note:







fog : Need not be defined if there exist any value in Range of function g for which function f is not defined then fog is not defined

In given situation, fog is defined then Range of g ⊆ domain of f

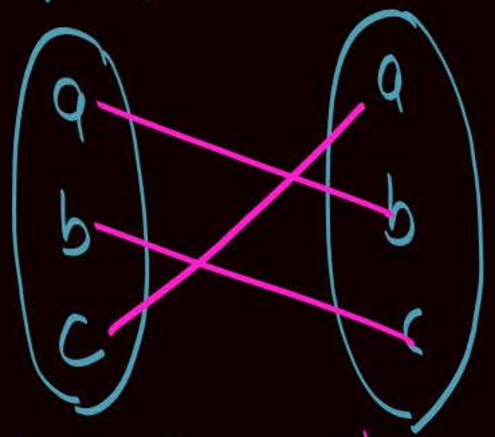
Note:

① Let $f: A \to A$ is a bijective function

$$f \circ f^{-1} = I_A$$

Similarly $f^{-1} \circ f = I_A$

A $f: A \to A$ A



$f(a) = b \Rightarrow f^{-1}(b) = a$

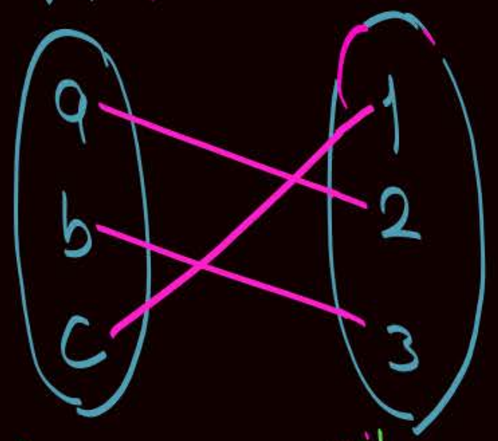
$(f \circ f^{-1})(b) = f(f^{-1}(b)) = f(a) = b$

② Let $f: A \to B$ is a bijective function

$$f \circ f^{-1} = I_B$$

whereas, $f^{-1} \circ f = I_A$

A $f: A \to B$ B



$f(a) = 2 \Rightarrow f^{-1}(2) = a$

$(f \circ f^{-1})(2) = f(f^{-1}(2)) = f(a) = 2$

Note: ① $f: A \to A$

$$f \circ I_A = f$$

$$(f \circ I_A)_{(x)} = f(I_{A(x)})$$
$$= f(x)$$
$$= f$$

Similarly

$$I_A \circ f = f$$

$$I_A(f(x)) = f(x)$$

② Let $f: A \to B$

$$f \circ I_A = f$$

$I_A \circ f$ = Need not be defined

$f \circ I_B$ = Need not be defined

$$I_B \circ f = f$$

Note:
IMP!

If '$*$' is any binary operation, then

$(x * y)^{-1} = y^{-1} * x^{-1}$ will always hold true

in general, $(x*y)^{-1} \neq x^{-1} * y^{-1}$;

$(x * y)^{-1} = x^{-1} * y^{-1}$ if '$*$' is a Commutative operation

$(fog)^{-1} = g^{-1} o f^{-1}$ (Always true)

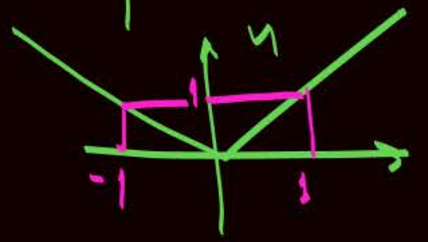
$(fog)^{-1}$ may or may not be equal to $(f^{-1} o g^{-1})$ { Because function Composition is not Commutative }

Q: Which of the following is/are a function on set of all real numbers

(a) $f(x) = \frac{1}{x}$ — $\frac{1}{x}$ is not defined at $x = 0$

(b) $f(x) = \sqrt{x}$ — $\sqrt{x}$ is not defined for '-ve' real numbers

(c) $f(x) = |x|$ — it is a function



{ but Not one-one
{ Not onto bcoz does not cover -ve real no.s }

(d) $f(x) = \pm\sqrt{x^2+1}$ — Each element of domain will have two images +ve & -ve ∴ Not a function

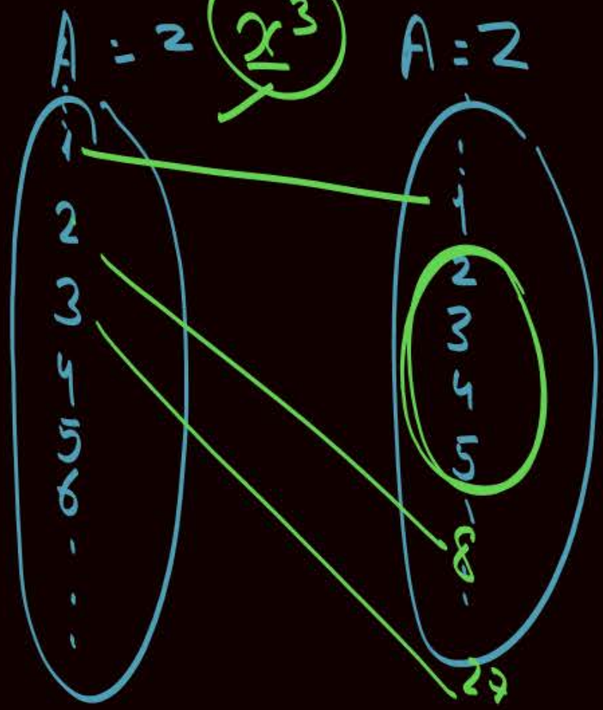
Q: Let f & g are two functions on set of all integers

$$f(x) = x^3 \qquad g(x) = \left\lceil \frac{x}{2} \right\rceil$$

$\left\lceil \frac{2}{2} \right\rceil = 1$ $\left\lceil \frac{1}{2} \right\rceil = 1$

Which of the given statements is/are true

- (a) $f(x)$ is one-one
- (b) $f(x)$ is onto {'2' is not a cube of any integer}
- (c) $g(x)$ is one-one
- (d) $g(x)$ is onto



Q. find the domain of function.

$$\frac{1}{\sqrt{|x|-x}}$$

Domain = $(-\infty, 0)$

$[-\infty, 0)$

for domain of given function

$$\sqrt{|x|-x} \neq 0$$

$$|x|-x \neq 0$$

$|x| = -x$ if $x < 0$

$|x| = x$ if $x \geq 0$

at $x < 0 \Rightarrow -x-x \neq 0$

$-2x \neq 0$

at $x \geq 0 \Rightarrow x-x = 0$



2 mins Summary

Topic Inverse of a function

Topic Identity function

Topic Constant function

Topic Identical function

Topic Function composition

Slide

THANK - YOU